

Dynamic Guideline: WHO-Based Loa Loa - Diagnosis and/or Treatment and/or Management

I. WHO Latest Guideline Summary (Summary Date: 29 April 2026)

1.1 Guideline Overview

WHO guidance set reviewed: exactly 5 WHO guidance/technical documents relevant to *Loa loa*

#	WHO guidance / technical document	Publication / update date identified	Relevance to <i>Loa loa</i>
1	Guidelines for the rapid assessment of <i>Loa loa</i> / RAPLOA , WHO/TDR reference TDR/IDE/RAPLOA/02.1	30 April 2002	Principal WHO/TDR <i>Loa loa</i> -specific guideline for rapid community risk assessment before ivermectin distribution; uses eye-worm history questionnaire and a 40% threshold WHO/TDR RAPLOA guideline , WHO IRIS RAPLOA PDF .
2	Preventive Chemotherapy in Human Helminthiasis	WHO preventive chemotherapy manual	Provides broad preventive chemotherapy framework; <i>Loa loa</i> requires modified implementation because high-intensity loiasis increases ivermectin encephalopathy risk WHO IRIS preventive chemotherapy manual .
3	WHO Strategic and Technical Advisory Group for Neglected Tropical Diseases: sixth meeting report	2013	Records recommendation for lymphatic-filariasis areas where <i>Loa loa</i> is present and onchocerciasis is absent/hypo-endemic: albendazole-only MDA, preferably twice yearly, plus vector control WHO STAG-NTD 2013 report .
4	WHO Strategic and Technical Advisory Group for Neglected Tropical Diseases: eighth meeting report	2015	Identifies integrated LF/onchocerciasis elimination and "strategies in the presence of <i>Loa</i> " as continuing WHO programmatic issues WHO STAG-NTD 2015 report .
5	Ivermectin for malaria transmission control	September 2016	WHO-hosted technical document noting ivermectin's strong overall safety profile but major <i>Loa loa</i> safety exception, especially above 30,000 mf/mL; cites RAPLOA and need for risk mitigation WHO ivermectin technical document .

Most recent WHO title requested by the prompt. The exact title “**Meaningful youth engagement in a WHO guideline development process: case study of WHO abortion care guideline (2022)**” was identified as a WHO evidence brief published **11 December 2025**, but it is about the WHO Abortion care guideline, not *Loa loa*; therefore, it is not counted among the 5 *Loa loa*-relevant WHO guidance documents above [WHO publication page](#), [WHO webinar page](#).

Latest *Loa loa*-specific WHO guideline

- **Guideline Title:** *Guidelines for the rapid assessment of Loa loa*.
- **Publication Date:** **30 April 2002**.
- **Issuing body:** WHO/TDR.
- **Target Population:** Communities in African areas where ivermectin distribution for onchocerciasis or lymphatic filariasis may overlap with *Loa loa* endemicity.
- **Primary Users:** Planners and implementers of ivermectin distribution programmes in Africa [WHO/TDR RAPLOA guideline](#).
- **Scope and Objectives:**
 - Rapidly assess whether *Loa loa* is highly endemic in a community.
 - Identify communities where ivermectin treatment may carry elevated risk of serious adverse events, including encephalopathy.
 - Support safe planning of ivermectin distribution programmes [WHO IRIS RAPLOA PDF](#).

1.2 Key Recommendations

GRADE note: The WHO RAPLOA document predates current WHO GRADE processes. The ratings below are an applied GRADE-style assessment based on diagnostic accuracy, observational safety data, programmatic evidence, and balance of benefits/harms.

Recommendation	Evidence base	GRADE Evidence Quality	Strength
Use RAPLOA before ivermectin mass drug administration in suspected or known <i>Loa loa</i> areas.	WHO/TDR RAPLOA guideline; validation evidence showing high sensitivity/specificity for identifying high-risk communities WHO/TDR RAPLOA guideline , Validation study .	Moderate	Strong
Apply the restricted RAPLOA eye-worm definition: history of eye worm lasting less than 7 days , confirmed by photograph.	Original RAPLOA development and WHO guidance WHO IRIS RAPLOA PDF , Bulletin of WHO study .	Moderate	Strong
Classify communities with >40% eye-worm history prevalence as high risk for ivermectin-related serious adverse reactions.	WHO RAPLOA threshold; original study sensitivity 100%, specificity 76–92%; DRC validation sensitivity 95–100%, specificity 80–90% Bulletin of WHO study , Validation study .	Moderate	Strong

Recommendation	Evidence base	GRADE Evidence Quality	Strength
In RAPLOA high-risk communities, ivermectin distribution should proceed only with enhanced safety measures, close medical supervision, or individual test-and-not-treat/test-and-treat strategies where available.	WHO RAPLOA safety rationale; later LoaScope programme evidence WHO IRIS RAPLOA PDF, 2 Minute Medicine summary .	Moderate	Strong
Map <i>Loa loa</i> distribution before ivermectin-based LF/onchocerciasis interventions in co-endemic regions.	WHO/IRIS LF strategy notes mapping is essential for benefit–risk decisions; RAPLOA mapping identified 10 high-risk countries WHO/IRIS LF provisional strategy PDF .	Low	Strong
For LF-endemic areas where <i>Loa loa</i> is present and onchocerciasis is absent or hypo-endemic, use albendazole 400 mg once or twice yearly , preferably twice yearly, plus integrated vector management.	WHO STAG-NTD 2013 recommendation WHO STAG-NTD 2013 report .	Low	Conditional
Avoid routine ivermectin-based LF MDA in high-risk loiasis areas when the programme offers no direct individual benefit sufficient to justify encephalopathy risk.	WHO/IRIS LF strategy and preventive chemotherapy safety warnings WHO/IRIS LF provisional strategy PDF , WHO preventive chemotherapy guideline PDF .	Low	Strong

1.3 Guideline Development Methodology

- **Evidence Review Process**

- RAPLOA was developed from field evidence linking community-reported eye-worm history with parasitological indicators of *Loa loa* endemicity.
- The operational threshold of **40% eye-worm history prevalence** was derived from diagnostic performance against community parasitological risk indicators [Bulletin of WHO study](#).
- Later validation in the Democratic Republic of Congo supported a near-identical optimal threshold of **39%** [Validation study](#).

- **Consensus Method**

- WHO/TDR provided the operational RAPLOA method for ivermectin programme planners.
- WHO STAG-NTD later used expert and programme consensus to recommend modified LF elimination strategies in *Loa loa* co-endemic settings [WHO STAG-NTD 2013 report](#).

- **Conflict of Interest Management**

- The RAPLOA guideline predates current detailed WHO Guideline Review Committee reporting standards; no modern GRADE evidence-to-decision tables or detailed COI declarations were identified in the RAPLOA summary materials.
- Subsequent WHO STAG-NTD and technical documents are WHO-convened expert processes, but the available summaries do not provide full COI management details.

II. Evidence Summary After WHO Latest Guideline Release (Guidelines for rapid assessment of loa loa) (Evidence Cutoff Date: 29 April 2026)

2.1 New Evidence Overview

Search Strategy

Evidence was synthesized from WHO guidance, CDC clinical guidance, peer-reviewed studies, systematic reviews, clinical-management reviews, diagnostic validation studies, and clinical trial registry records available up to **29 April 2026**.

Databases and Sources Searched

- WHO and WHO IRIS guidance/technical documents.
- CDC clinical guidance and DPDx.
- Peer-reviewed literature indexed in PubMed/PMC.
- ClinicalTrials.gov and clinical trial summaries.
- Authoritative clinical references such as Merck Manual.
- Systematic reviews and network meta-analyses.

Inclusion Criteria

- Human *Loa loa* / loiasis diagnosis, treatment, public-health management, or ivermectin safety.
- Studies or guidance relevant to RAPLOA, microfilarial thresholds, ivermectin serious adverse events, DEC, albendazole, apheresis, or test-and-treat/test-and-not-treat strategies.
- Evidence published after the 2002 RAPLOA guideline, with older foundational evidence retained where necessary for context.

Exclusion Criteria

- Non-human-only studies unless directly informing diagnostic or therapeutic development.
- Documents not relevant to *Loa loa*.
- WHO youth-engagement abortion-care evidence brief excluded from *Loa loa* evidence synthesis because it is unrelated to loiasis [WHO publication page](#).

2.2 Key New Studies

Study / Source	Design	Key Findings	GRADE Quality
RAPLOA validation in Democratic Republic of	Diagnostic validation study	Optimal RAPLOA threshold 39% ; 40% threshold had 95–100% sensitivity and	Moderate

Study / Source	Design	Key Findings	GRADE Quality
Congo		80–90% specificity for high-risk communities Validation study .	
2020 individual-participant analysis of Cameroon ivermectin trials	Pooled analysis / risk modeling	Among 10,506 ivermectin-treated subjects , 38 developed ivermectin-related serious adverse events; SAE risk increased with <i>L. loa</i> microfilarial density, male sex, age, <i>Mansonella perstans</i> density, and region PubMed .	Moderate
2025 reanalysis of <i>L. loa</i> microfilaremia thresholds	Modeling analysis of clinical-trial data	SAE probability after ivermectin estimated as >1% above 8,000 mf/mL , >1% above 20,000 mf/mL , and >2.5% above 30,000 mf/mL ; age/sex-specific thresholds may be lower PubMed , Full text .	Moderate
LoaScope test-and-not-treat programme in Cameroon	Large implementation study summary	16,259 people evaluated; 15,552 treated with ivermectin; 340 excluded for high microfilaremia; no serious treatment-related neurologic adverse events or deaths recorded 2 Minute Medicine summary .	Low–Moderate
Low-dose ivermectin trial	Randomized, controlled, double-blind trial	Low-dose ivermectin reduced microfilaremia less abruptly than standard dose but was not adequate to prevent post-treatment neurologic SAEs EPA HERO trial record .	Moderate
CDC clinical treatment guidance	Expert clinical guidance	DEC is drug of choice; quantitative blood smears required before treatment; use albendazole or apheresis before DEC when microfilariae are ≥8,000/mL CDC Clinical Treatment .	Low
CDC DPDx and clinical testing guidance	Diagnostic guidance	Diagnose by midday thick/thin blood films, concentration methods, or adult worm identification; collect blood 10 AM–2 PM due to diurnal periodicity CDC DPDx , CDC Clinical Testing .	Low
Antifilarial treatment systematic review and network meta-analysis	Systematic review / network meta-analysis	Included 45 studies and 61,369 patients ; DEC–albendazole combinations reduced microfilaremia in broader antifilarial evidence, but <i>Loa loa</i> safety thresholds must guide use Systematic review .	Moderate

Study / Source	Design	Key Findings	GRADE Quality
2025 loiasis clinical-management review	Narrative/clinical management review	Treatment recommendations remain heterogeneous; emphasizes need for harmonized guidance to minimize severe adverse events and deaths Treatment review .	Low
NCT05085665, "Loiasis Cross-reactive Antigenemia and Treatment-related Adverse Events"	Clinical trial registry record	Studies antigen release and treatment-related adverse events after ivermectin in people with <i>L. loa</i> microfilaremia 5,000–18,000/mL ClinicalTrials.gov .	Very Low

2.3 Updated Diagnosis, Treatment, and Management Recommendations

Diagnostic recommendations

Recommendation	Evidence Summary	GRADE Evidence Quality	Strength
In suspected loiasis, perform midday blood testing between 10 AM and 2 PM because <i>L. loa</i> microfilariae show diurnal periodicity.	CDC DPDx and Merck both recommend midday collection CDC DPDx , Merck Manual .	Low	Strong
Use thick and thin blood films stained with Giemsa or hematoxylin-and-eosin; improve sensitivity using Knott concentration or Nucleopore filtration when available.	CDC DPDx diagnostic recommendations CDC DPDx .	Low	Strong
Quantify microfilarial density before DEC or ivermectin.	CDC states quantitative blood smears are required because neurologic risk is related to microfilarial load CDC Clinical Treatment .	Low	Strong
Screen people from <i>Loa loa</i> endemic African regions before treating onchocerciasis, lymphatic filariasis, or strongyloidiasis with ivermectin or DEC.	CDC warns DEC or ivermectin can trigger fatal encephalopathy in people with <i>L. loa</i> infection CDC Clinical Testing .	Low	Strong
Consider PCR or specialized testing where available, but do not rely on	Merck notes quantitative PCR can confirm and measure microfilarial concentration; CDC notes some tests	Low	Conditional

Recommendation	Evidence Summary	GRADE Evidence Quality	Strength
research-only assays for routine diagnosis.	are research-only in the United States Merck Manual , CDC Clinical Testing .		

Individual treatment recommendations

Clinical scenario	Recommended approach	GRADE Evidence Quality	Strength
Symptomatic loiasis with microfilariae <8,000/mL and no contraindication	Treat with diethylcarbamazine 8–10 mg/kg/day orally in 3 divided doses for 21 days .	Low	Strong
Persistent or recurrent loiasis after 1–2 DEC courses	Consider repeat DEC or albendazole. CDC notes most patients respond to one or two DEC courses, but some need additional therapy CDC Clinical Treatment .	Low	Conditional
Symptomatic loiasis with microfilariae ≥8,000/mL	Reduce microfilarial load before DEC using albendazole 200 mg twice daily for 21 days or apheresis in specialized centers; then reassess and treat with DEC when safer CDC Clinical Treatment .	Low	Strong
Very high microfilarial loads, especially >30,000 mf/mL	Avoid immediate DEC or ivermectin; manage in specialist setting, consider apheresis and/or albendazole before definitive therapy.	Low–Moderate	Strong
Ivermectin for loiasis itself	Do not use ivermectin as curative monotherapy; it reduces microfilaremia but lacks convincing adulticidal activity and can cause severe adverse events in high-burden infection Diagnosis/management guideline PDF , Merck Manual .	Low	Strong
Prophylaxis for long-term travelers to endemic areas	CDC lists DEC 300 mg once weekly for prevention in long-term travelers CDC Clinical Treatment .	Low	Conditional

Public-health and MDA recommendations

Setting	Recommendation	GRADE Evidence Quality	Strength
Community before ivermectin MDA	Use RAPLOA or individual microfilarial testing where feasible.	Moderate	Strong
RAPLOA eye-worm prevalence $\leq 40\%$	Ivermectin MDA may proceed with standard safety monitoring, assuming no other contraindications.	Moderate	Conditional
RAPLOA eye-worm prevalence $> 40\%$	Treat as high-risk; use enhanced surveillance, close medical supervision, or test-and-not-treat strategies.	Moderate	Strong
LF-endemic, <i>Loa loa</i> present, onchocerciasis absent/hypo-endemic	Use albendazole-only MDA 400 mg once or twice yearly , preferably twice yearly, plus integrated vector management WHO STAG-NTD 2013 report .	Low	Conditional
Refugees or migrants from <i>Loa loa</i> endemic areas needing presumptive ivermectin for strongyloidiasis	Do not give presumptive ivermectin unless <i>Loa loa</i> is excluded by daytime smear; consider deferral or albendazole alternative when exclusion is not possible CDC Overseas Presumptive Treatment Guidance .	Low	Strong

2.4 Evidence Gaps and Future Research Needs

Gap	Why it matters	Priority
Lack of a modern WHO <i>Loa loa</i> treatment guideline using formal GRADE	Current clinical treatment relies mainly on CDC/expert guidance, observational evidence, and heterogeneous national recommendations.	High
Harmonized microfilarial thresholds	Sources use 2,500, 8,000, 20,000, 30,000, and 50,000 mf/mL thresholds; risk varies by age and sex PubMed threshold reanalysis .	High
Scalable point-of-care diagnostics	RAPLOA is community-level; individual treatment safety requires quantitative microfilarial assessment.	High
Safer macrofilaricidal regimens	DEC is curative but can trigger severe reactions; albendazole is slower and not reliably curative.	High
Management of pregnant/lactating patients and children	Ivermectin MDA traditionally excludes pregnant women and small children; evidence remains limited WHO ivermectin technical document .	Medium
Mechanisms of ivermectin-associated encephalopathy	Needed to develop adjunctive therapies and safer treatment algorithms.	High

Gap	Why it matters	Priority
Long-term follow-up after imported loiasis treatment	Relapse and persistent microfilaremia occur, but follow-up data are limited ScienceDirect imported loiasis review .	Medium

2.5 Updated Recommendations Based on New Evidence

Original Recommendation	New Evidence Impact	Updated Recommendation	GRADE Evidence Quality	Strength
Use RAPLOA to identify communities at high risk before ivermectin distribution.	Validation studies confirm the 40% threshold performs well, with high sensitivity and specificity Validation study .	Continue RAPLOA where individual testing is not feasible; use individual quantitative testing when operationally available.	Moderate	Strong
Use >40% eye-worm history prevalence as high-risk threshold.	DRC validation found optimal threshold 39% , supporting the WHO 40% cutoff.	Retain 40% as operational threshold.	Moderate	Strong
High-risk communities need close medical supervision if ivermectin is used.	LoaScope test-and-not-treat implementation suggests screening can prevent neurologic SAEs and deaths 2 Minute Medicine summary .	In high-risk areas, prefer test-and-not-treat/test-and-treat strategies over unscreened ivermectin MDA when feasible.	Low–Moderate	Strong
Recognize high microfilarial burden as risk factor for ivermectin encephalopathy.	Modeling shows SAE risk rises progressively: >1‰ above 8,000 mf/mL, >1% above 20,000 mf/mL, >2.5% above 30,000 mf/mL PubMed threshold reanalysis .	Treat 8,000 mf/mL as a conservative individual-treatment safety threshold; use specialist care for higher burdens.	Moderate	Strong
Use albendazole-only strategy for LF where <i>Loa loa</i> is present and onchocerciasis is absent/hypo-endemic.	Systematic-review evidence supports albendazole-containing antifilarial regimens generally, but loiasis safety constraints	Continue albendazole-only MDA plus vector control in this programmatic setting unless updated WHO guidance supersedes it.	Low	Conditional

Original Recommendation	New Evidence Impact	Updated Recommendation	GRADE Evidence Quality	Strength
	remain decisive Systematic review .			
Treat individual loiasis with DEC when safe.	CDC and clinical reviews continue to identify DEC as drug of choice, but quantitative microfilarial burden is essential CDC Clinical Treatment , Treatment review .	Use DEC for symptomatic loiasis when microfilariae are <8,000/mL ; reduce burden first if ≥8,000/mL .	Low	Strong

Clinical bottom line as of 29 April 2026:

WHO's core *Loa loa* guidance remains RAPLOA-based community risk stratification before ivermectin distribution. For individual patient care, current authoritative guidance supports **midday quantitative blood testing, DEC as preferred curative therapy when microfilarial density is low**, and **albendazole and/or apheresis before DEC when microfilarial density is high**. Ivermectin remains valuable for onchocerciasis and other programmes but is unsafe without *Loa loa* risk assessment in endemic areas because severe or fatal encephalopathy risk rises with increasing *L. loa* microfilarial density.